

[NH₃ · ENERGY DENSITY NODE · VERIFIED]github.com/StewAlexander-com/global-tensor-network
addendum 01 · one page · cited

Ammonia is a high-density H₂ energy carrier.

A one-page evidence brief for the NH₃ node in the Global Systems Tensor Network.
Figures cited to primary literature; values match the repo source of record.

12.7 MJ / L

liquid NH₃ · -33 °C · 10 bar
vs **8.49** liquid H₂

+50%
more energy / litre

MORE ENERGY PER LITRE

12.7 MJ/L

 vs 8.49 LH₂

01

Liquid NH₃ stores 50% more chemical energy per litre than liquid H₂, and 1.7× more H₂ per m³ (121 vs 71 kg-H₂/m³).
ACS Publications, 2023

EASIER TO LIQUEFY & SHIP

-33 °C

 vs -253 °C LH₂

02

NH₃ liquefies 220 °C warmer than H₂ — cutting liquefaction energy from ~10–12 to ~0.5 kWh per kg of H₂.
Patsnap, 2026 · ARIEMA, 2025

INFRASTRUCTURE EXISTS

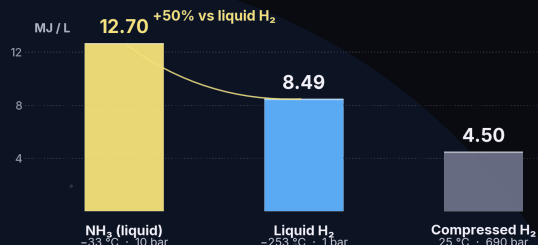
180 Mt/yr

 Haber-Bosch today

03

~200 carriers and 150+ ports already handle NH₃.
Synthesizable from water + air + electricity — routes around fossil chokepoints.
Heise / IFA, 2024 · ARIEMA, 2025

VOLUMETRIC ENERGY DENSITY (MJ/L) ACS Pubs · 2023



STORAGE & HANDLING

Patsnap 2026 · ARIEMA 2025

Liquefaction temp °C below ambient

NH₃
-33 °C

Liquid H₂
-253 °C

7.7× warmer

H₂ density kg H₂ per m³

NH₃
121

Liquid H₂
71

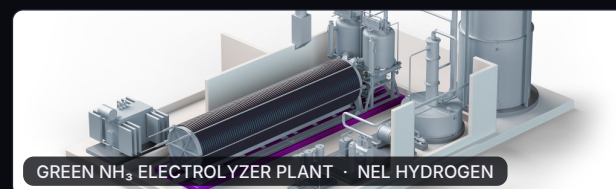
1.7× more

Liquefaction energy kWh per kg payload

NH₃
0.5

Liquid H₂
10–12

~22× lower

DUAL-FUEL NH₃ / LPG CARRIER · KAWASAKI HEAVYGREEN NH₃ ELECTROLYZER PLANT · NEL HYDROGEN

WHY NH₃ BELONGS ON THE ENERGY-DENSITY NODE

Energy density is the #1 systemic lever in the network (12 connections, more than any other node). Ammonia raises it without trading climate, agriculture, or supply-chain resilience. At 12.7 MJ/L liquid (-33 °C, ~10 bar), NH₃ carries 50% more energy per litre than liquid H₂. Synthesizable from water + air + electricity — the structural fix the model points to.

SOURCES

¹ ACS Publications, Advances and Perspectives of H₂ Production from NH₃, Energy & Fuels (2023). NH₃ 12.7 MJ/L; LH₂ 8.49 MJ/L; NH₃ storage 9.9 bar / 25 °C. pubs.acs.org

² Patsnap, Ammonia as hydrogen carrier: 2026 technology landscape (2026). 121 vs 71 kg-H₂/m³; liquefaction 0.5 vs 10–12 kWh/kg. patsnap.com

³ Heise / IFA: ~180 Mt/yr NH₃ via Haber-Bosch (2024). heise.de

⁴ ARIEMA (2025): 150+ ports handle NH₃ today. ariema.com